

Customer Credit Risk: Application and Evaluation of Machine Learning and Deep Learning Models

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Abstract - Financial institutions rely on credit risk evaluation or credit scoring models for decision-making, as these models analyse the creditworthiness of customers. Machine learning models can predict with high accuracy, and they are widely utilized in this sector. This study selected a dataset from openml.org, and after data pre-processing, visualization, and exploratory data analysis, seven algorithms were applied. The models were evaluated using various performance metrics. Support Vector Machines (SVM) achieved the highest accuracy (80.67%) with a good Recall of 93.55%. Also, when dimensions were reduced to 10 (from 20) through Principal Component Analysis, SVM demonstrated the highest accuracy (83.57%), an F1 score of 84.70%, a Recall of 88.84%, and an ROC AUC Score of 83.44%. By generating synthetic records using SMOTE, the open-source algorithm Extreme Gradient Boosting achieved the highest accuracy score of 83.3% and an ROC AUC of 83.29%. Future work may involve tuning the hyperparameters of these algorithms to improve other performance metrics.

Keywords – Customer classification, ML algorithms, credit scoring

I. INTRODUCTION

Credit risk evaluation or credit scoring is crucial for any financial institution. It is extensively utilized by financial institutions to analyze creditworthiness [1] before extending any debts to customers. Misclassifying customers into incorrect categories can result in bad credits and financial losses. With strict regulations imposed on financial institutions, customer classification models must be accurate as these institutions base decisions on the output of these models, and they are significantly impacted by misclassification. Numerous contributing factors such as age, existing loans, credit amount, marital status and others make the models complex.

There are many traditional statistical models used for credit risk evaluation since decades. These models are also called as ‘credit scoring’ models. These traditional models have been using the methods like Discriminant Analysis (DA) and Logistic Regression (LR) and they can classify with high

accuracy. Machine learning has slowly entered this segment and algorithms like Artificial Neural Networks (ANN), Deep Learning algorithms and Support Vector Machines (SVM) are being widely used to classify customer credits.

There have been numerous studies that have applied Machine Learning and Deep Learning algorithms to credit risk evaluation models. Azaria et al. [1] utilized various non-parametric and parametric statistics to compare different Machine Learning classifiers used on customer loans. The Deep Neural Network achieved the highest Area Under the Curve (AUC) of 0.638.

In other studies [2], Bayesian, Random Forests (RF), SVM, and Naïve Bayes classifiers were applied to publicly available German credit scoring data, with a comparative study conducted. Different feature selection algorithms such as Information-Gain, Gain-Ratio, and Chi-Square were employed in conjunction with classifiers. Random Forests and Chi-Square provided the highest accuracy, and the result of the Decision Tree (C5.0) with Chi-Square was also noteworthy.

Louis et al. [3] applied Machine Learning and Deep Learning methods to highly unbalanced 24-month credit card data. AutoML and Neural Architecture Search (NAS) techniques were utilized, with the Extreme Gradient Boosting (XGBoost) algorithm achieving the highest AUC of 0.78. Logistic Regression, Decision Trees, and Random Forest algorithms were applied to microfinance institution data in [4], with RF achieving 83% accuracy. Ping et al. [5] employed Bagging Classification and Regression Trees (CART) and AdaBoost CART on Australian Credit Card data from the UCI dataset with 14 features, and AdaBoost CART achieved the highest accuracy of 85.86%.

I Nyoman et al. [6] applied an ensemble model of XGBoost and Random Forests to telecom and insurance customer data and compared the results with those of similar studies. This model achieved better F1 scores (0.85 for telecom data and

0.947 for insurance data) and AUC scores (0.857 and 0.99 for telecom and insurance data respectively) compared to other studies. In other studies [7], Random Forests, Decision Tree, Logistic Regression, and AdaBoost algorithms were applied to a Kaggle credit card dataset, with RF achieving the highest accuracy of 94.4% in identifying frauds.

Bing et al. [8] proposed a hybrid Deep Learning model that combined Convolutional Neural Network (CNN) and Relief algorithm. They applied this hybrid model to consumer credit data collected from a Chinese finance company and compared it with Random Forest and Logistic Regression models. Relief-CNN achieved an accuracy of 91.64% and yielded superior results compared to the other two models.

This study applies the traditional Logistic Regression algorithm along with the powerful Extreme Gradient Boosting open-source library. Seven different ML algorithms are utilized on this dataset. These include Conventional Logistic Regression (Maximum Entropy Based) [9], Multi-Layer Perceptron (Neural Network), along with Boosting algorithms such as Extreme Gradient Boosting and Random Forest Classifier (Ensemble Based-Bagging), Decision Tree Classifier (Tree Based), K Nearest Neighbors (Group Based), and Support Vector Machines. Performance metrics are utilized to compare the models.

II. METHODOLOGY

This study methodology is classified into three phases.

Phase 1. Data Processing

Phase 2. Application of Machine Learning Algorithms

Phase 3. Performance metrics and Model evaluation

Each phase is explained in detail below.

A. Phase I – Data processing

There are many open-source datasets available on internet for customer credit risk evaluation. The dataset chosen for this study is credit_customers.csv taken from openml.org [10]. There are 1000 records in the dataset with 21 features. Seven features are float datatype, and 14 features are object datatype. Feature 'class' is the target feature that holds value 'good' for good customers who can be given loans and 'bad' for customers who can put the financial institutions in risk. checking_status, other_parties, housing, other_payment_plans, residence_since, job, duration, age, credit_history, purpose, credit_amount, savings_status, employment, foreign_worker, installment_commitment, own_telephone, personal_status, num_dependents, property_magnitude, existing_credits and class are the features in the dataset.

Data Pre-processing To understand the features and their relationships, Exploratory Data Analysis (EDA) is conducted on the dataset. No duplicates or missing values are found. The

data is checked for outliers, and as the outliers did not appear to be true outliers, they are not treated. Two new features, 'gender' and 'marital status', extracted from 'personal status', appear to be highly correlated (0.94). To avoid multicollinearity, they are removed. One-hot encoding using Pandas.get_dummies is applied for 14 categorical features. The target feature 'class' is converted into numerical values. Standard Scaler is used to scale the data.

Data Partitioning Training data and Test data are obtained from the dataset by splitting it into 70:30 ratio. 700 are in training set and test set has 300 records.

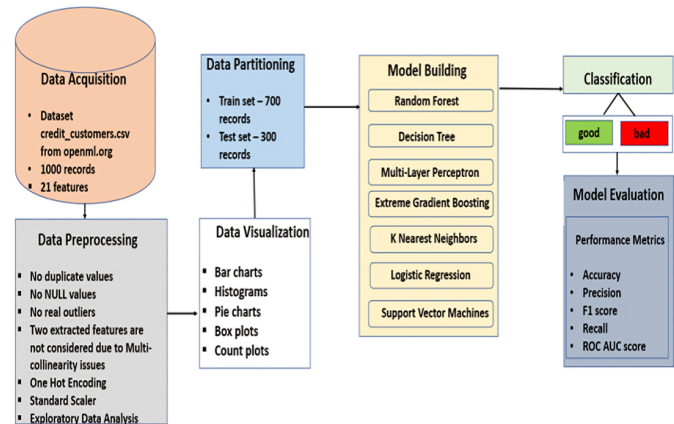


Fig. 1 Methodology of this study

B. Phase II – Application of ML algorithms

Seven different algorithms are chosen for this study. Logistic Regression, Multi-Layer Perceptron (MLP), Extreme Gradient Boosting, Random Forest Classifier, Decision Tree Classifier, K Nearest Neighbors and Support Vector Machines are applied to the dataset. Performance metrics are used to compare the models. Details of the algorithms are explained below.

Logistic Regression – This is a traditional algorithm that is used for classification, and it predicts the target class using probability. Though it is used for classification, it is still called as regression [11] as it uses the output from linear regression function and applies a sigmoid function to identify the output of a target class.

Multi-Layer Perceptron – MLP comes under feedforward algorithms of Artificial Neural Network that has input, hidden and output layers. Non-linear mapping [12] exists between input and output in this algorithm.

Extreme Gradient Boosting – This is a powerful open-source library that implements gradient boosting. Extreme Gradient Boosting algorithm won [13] many Kaggle competitions for its accuracy in predictions.

Random Forests - This is a supervised learning algorithm that contains many decision trees. The outcome of this

algorithm is based on the predictions [14] made by the decision trees.

Decision Tree – Decision Tree is a non-parametric algorithm [15] that can handle multi-output problems and uses decision rules to predict the output.

K-Nearest Neighbors – KNN is an extremely easy algorithm to implement. It uses proximity [16] to decide the outcome.

Support Vector Machines – SVM is a powerful algorithm that generates the optimal hyperplane [17] to differentiate the target classes.

C. Phase III – Evaluation metrics

Evaluation metrics used in this study are given below. Five metrics - Accuracy, F1, Receiver Operating Characteristic Area Under the Curve (ROC AUC), Recall and Precision scored are used to evaluate the models.

$$Accuracy = \left\{ \frac{(TP + TN)}{(TP + FP + TN + FN)} \right\} \in \{0,1\} \quad (1)$$

$$Precision\ Score = \left\{ \frac{TP}{TP + FP} \right\} \quad (2)$$

$$Recall\ Score = \left\{ \frac{TP}{TP + FN} \right\} \quad (3)$$

$$F1 = \frac{2 \times P \times R}{P + R} \in \{0,1\} \quad (4)$$

$$ROC\ AUC = \int_0^1 \left(\frac{TP}{TP + TN} \right) d \left(\frac{FP}{FP + FN} \right) = \int_0^1 \left(\frac{TP}{P} \right) d \left(\frac{FP}{N} \right) \quad (5)$$

Accuracy (1) is the ratio of correct predictions to total number of predictions made. TP = True positives, TN = True negatives, FP = False positives and FN = False negatives.

Precision is the ability of the model to predict positive labels. A model is expected to have this score closer to one.

Recall score defines how well a model can classify the positive class when the outcome is positive.

F1 score is a harmonic mean of Precision and Recall.

ROC AUC score indicates how effectively the classification algorithm can differentiate between positive and negative classes. The higher the ROC AUC score, the better the performance of the model.

Experimental setup Dell Inspiron 7506 laptop is used to perform the analysis and it has a 16 GB Intel Core i7 processor and a 512 GB SSD. Jupyter Notebook and Python is used for this study. Pandas, Numpy, Matplotlib, Scikit-learn, Seaborn libraries are used.

III. RESULTS

A. Data Visualization

Bar charts, Histograms, Pie charts, Box plots and Count plots are used to visualize the data. Heatmap generated for the dataset is shown in Fig.2.

Features checking_status (positively correlated), purpose (positively correlated), savings_status (positively correlated), personal_status (positively correlated) and duration (negatively correlated) are the significant features in identifying customer class.

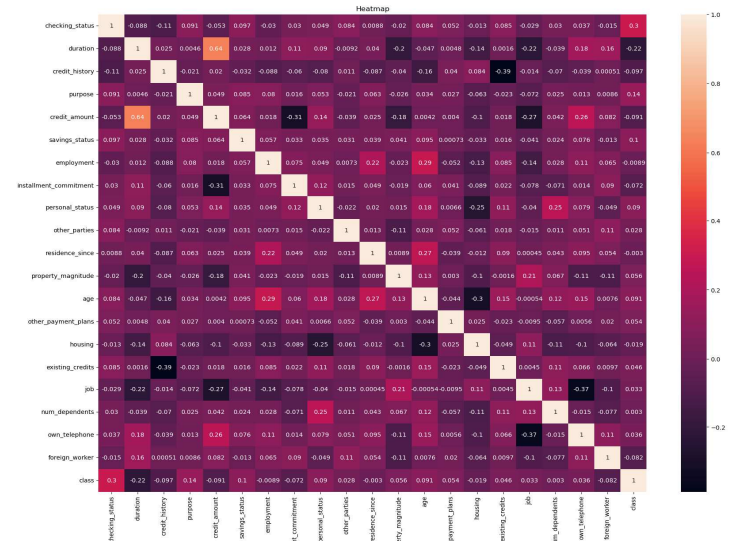


Fig.2 Heatmap generated for the dataset

B. Model Evaluation

Different evaluation metrics obtained for seven Machine Learning models applied to the dataset are presented in Table 1. The best evaluation metrics for different techniques are highlighted in blue in Table 1, 2 and 3.

The models performed better with one-hot encoding compared to Label encoding. Therefore, one-hot encoding is subsequently used in the dataset. With one-hot encoding, Support Vector Machines achieved the highest accuracy of 80.67%. This demonstrates that SVM is a powerful algorithm for distinguishing target classes in feature space, achieving a good recall of 93.55%.

Due to the data imbalance, Synthetic Minority Oversampling Technique (SMOTE) is applied to create

synthetic records and balance the data. With SMOTE, Extreme Gradient Boosting attained the highest accuracy score of 83.3% and an ROC AUC of 83.29%.

Principal Component Analysis is conducted on the data to reduce the dimensions from 21 to 10. With reduced dimensions, SVM achieved the highest accuracy (83.57%), an F1 score of 84.70%, a recall of 88.84%, and an ROC AUC Score of 83.44%.

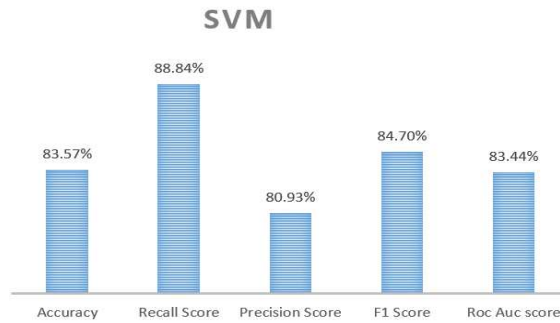


Fig.3 Evaluation metrics of SVM

Table.1 Accuracy and F1 scores of algorithms expressed in percentages

Algorithm	Accuracy				F1 score			
	Label Enc	1 Hot	SMOTE	PCA	Label Enc	1 Hot	SMOTE	PCA
RF	78.00	80.33	82.62	81.67	85.90	80.33	82.62	82.77
DT	67.67	74.33	77.14	74.76	76.63	87.31	83.45	75.00
MLP	75.00	77.67	83.10	83.10	83.07	82.13	77.03	83.75
XGB	78.33	78.33	83.33	80.95	85.39	84.81	83.29	81.90
KNN	74.00	76.33	80.00	80.48	83.19	85.26	83.95	81.11
LR	74.67	77.33	81.19	83.33	84.10	84.86	80.19	83.95
SVM	75.33	80.67	80.24	83.57	84.71	84.82	81.84	84.70

Table.2 Recall scores of algorithms expressed in percentages

Algorithm	Recall			
	Label	1 Hot	SMOTE	PCA
RF	92.63	93.55	85.58	86.05
DT	73.27	81.57	74.88	73.95
MLP	84.79	86.18	82.33	85.12
XGB	87.56	86.64	85.12	84.19
KNN	88.94	91.71	79.07	81.86
LR	92.63	87.56	82.79	85.12
SVM	94.47	93.55	86.51	88.84

Table.3 Precision and ROC AUC scores of algorithms expressed in percentages

Algorithm	Precision				ROC AUC			
	Label Enc	1 Hot	SMOTE	PCA	Label Enc	1 Hot	SMOTE	PCA
RF	80.08	81.85	81.42	79.74	66.19	69.67	82.55	81.56
DT	80.30	82.71	79.31	76.08	63.14	68.49	77.20	74.78
MLP	81.42	83.48	84.29	82.43	67.10	70.80	83.11	83.05
XGB	83.3	83.9	82.81	79.7	70.8	71.6	83.29	80.8

	3	3		4	9	3		7
KNN	78.14	78.97	81.34	80.37	61.94	63.92	80.02	80.44
LR	77.01	82.25	80.91	82.81	60.17	69.08	81.15	83.29
SVM	76.78	82.19	77.50	80.93	59.89	70.27	80.09	83.44

SVM has performed well when compared to other ML models and the evaluation metrics of SVM are shown in Fig. 3. However, lower ROC AUC score of 70.27% show that the algorithm must be tuned with the best hyperparameters to improve the prediction accuracy.

IV. DISCUSSION

Machine Learning has incredibly good algorithms for classification. Several important studies that used Machine Learning classification algorithms for customer credit risk evaluation are discussed in this study. Seven Machine Learning algorithms, including Deep Learning algorithm Multi-Layer Perceptron, Logistic Regression, boosting algorithms - Extreme Gradient Boosting and Random Forests, Decision Trees, K Nearest Neighbors, and Support Vector Machines are applied in this study on a publicly available dataset. Support Vector Machines achieved the highest accuracy with Extreme Gradient Boosting being the best for a well-balanced dataset. The results are summarized and presented in the tables.

V. CONCLUSION

Credit risk evaluation is important for financial institutions and the models used for credit scoring must be accurate in their predictions. In conclusion, this study has applied Machine Learning as well as Deep Learning algorithms to classify customer credit risk. The results showed that Support Vector Machines is the best for classification with the open-source Extreme Gradient Boosting algorithm performing well on a balanced dataset. In the future, we intend to tune the hyperparameters for these algorithms to improve the performance metrics.

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